

Aayush Yadav

Full Stack AI Developer

+91 7021907344 - aayushdyadav2003@gmail.com

github.com/03ADY -Portfolio

Professional Summary

Full Stack AI Developer with production experience designing and operating enterprise AI platforms spanning Retrieval-Augmented Generation (RAG), GenAI presales, inventory management, HR automation, support copilots, and workflow automation. Experienced in Python, TypeScript, React, FastAPI, AWS cloud services, LangChain, Gemini, scalable backend systems, and multi-tenant architectures. AWS Certified Solutions Architect with a proven record of delivering AI-powered systems used by enterprise teams in production.

Professional Experience

Full Stack AI Developer -Allwave (Remote)

Jan 2026 -Present

- Led development of six production AI platforms across inventory operations, GenAI presales, support automation, HR SaaS, AV programming, and 3D room design.
- Built scalable full-stack applications using TypeScript, React, Python, FastAPI, Gemini, LangChain, and AWS services.
- Designed cloud-native architectures utilizing Amplify Gen 2, Lambda, DynamoDB, Cognito, App Runner, AppSync, SES, and CloudWatch.
- Owned CI/CD pipelines, asynchronous LLM workflows, multi-tenant RBAC, monitoring, and production deployments.
- Automated India-specific compliance workflows including GST, MSME, TDS, and Tally XML exports.

AI Engineer -Jobihood Technologies (Remote)

Nov 2025 -Jan 2026

- Developed a Gemini-powered OCR Resume Intelligence platform converting PDFs into structured candidate profiles.
- Built multi-tenant vector search with semantic retrieval and LLM intent parsing.
- Designed enterprise-grade candidate ranking and seniority estimation pipelines.

Python Backend Developer Intern -NDTR Reflection Engineers (Remote)

Oct 2024 -Mar 2025

- Built backend microservices using Python, FastAPI, and Flask.
- Improved API response time by 25% while maintaining 99.9% uptime.
- Optimized PostgreSQL performance, secured JWT APIs, integrated SendGrid and Razorpay, and reduced deployment time using Docker.

Industry Projects

Production systems designed, built, and operated at Allwave.

AV Inventory Ops -Cloud-Native Operations Platform

TypeScript · Amplify Gen 2 · AppSync GraphQL · DynamoDB · Lambda · React PWA · Gemini · SES

- Designed and shipped a serverless operations platform that consolidated inventory, procurement, billing, AMC/service workflows, and client portal into a single system of record, replacing fragmented spreadsheet-driven processes.
- Architected a domain data model across 36 DynamoDB tables (8 GSIs) and 19 purpose-built Lambda functions covering invoicing, GST/MSME/TDS automation, FY rollover, alerting, and Tally XML export for finance and compliance handoff.
- Delivered a production React PWA with QR-based unit tracking, Cmd+K navigation, embedded Gemini assistant, WAF-protected endpoints, Secrets Manager, and point-in-time recovery for operational resilience.

BINGO -GenAI Presales & BOQ Generation Platform

TypeScript · Amplify Gen 2 · Gemini · LangChain RAG · DynamoDB · Cognito · Excel/PDF/DOCX

- Built an end-to-end GenAI presales platform that generates AVIXA CTS-D compliant Bills of Quantities from a curated catalog of 2,000+ AV products, reducing multi-day proposal cycles to hours.
- Implemented Ask BINGO RAG with hybrid retrieval (vector + database search), IndexedDB caching, and exact-model matching to ground LLM outputs in approved SKUs and minimize product hallucination.
- Established asynchronous Gemini job queues for large proposal runs, Cognito-based RBAC, audit logging, and multi-format client deliverables (Excel, PDF, Word).

PRONTO -Production RAG Support Copilot

Python · FastAPI · LangChain · FAISS · Gemini · Jira REST · Lambda · React

- Developed a production RAG support copilot grounded in internal knowledge bases to surface cited, context-aware resolutions for L1/L2 teams under live incident SLAs.
- Integrated Jira REST APIs to enrich tickets with AI analysis, structured solution commentary, and escalation signals, reducing manual triage effort on recurring issues.
- Shipped React analytics for SLA and priority visibility, with lazy-loaded RAG/Jira/escalation services engineered for AWS Lambda cold-start constraints.

HiRo -Multi-Tenant HR Automation SaaS

FastAPI · App Runner · Gemini · DynamoDB · Cognito · SES · CodeBuild · ECR

- Delivered a multi-tenant hiring and onboarding platform featuring a Gemini HR copilot, branded email templates, workflow-triggered sequences, and campaign-based outreach.
- Implemented tenant isolation with Cognito RBAC, SES domain verification for production deliverability, and auditable send/activity logs.
- Owned the release pipeline from GitHub to CodeBuild, ECR, and App Runner, with CloudWatch-based production monitoring and operational runbooks.

Personal Projects

Aura -Production Retrieval-Augmented Generation Platform

- Developed a production-grade RAG platform using FastAPI, Gemini 2.0, LangChain, ChromaDB, and Docker.
- Implemented history-aware retrieval, Maximum Marginal Relevance (MMR), session memory, document lifecycle APIs, citations, and scalable REST APIs.

RetainAI -Customer Churn Prediction

- Built an end-to-end customer churn prediction platform using XGBoost, feature engineering, model calibration, and Streamlit dashboards.
- Developed business ROI analysis tools to estimate retention campaign effectiveness and support data-driven business decisions.

Technical Skills

Languages: Python, TypeScript, SQL, JavaScript

Frameworks: React 18, FastAPI, Flask, Express, Node.js, Pydantic

Cloud & DevOps: AWS Amplify Gen 2, Lambda, DynamoDB, Cognito, App Runner, AppSync, SES, EventBridge, CloudWatch, CloudTrail, S3, ECR, CodeBuild, Docker, GitHub Actions, CI/CD

Generative AI: Gemini, LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, ChromaDB, FAISS, Vector Search, Hugging Face Transformers, Async LLM Workflows

Machine Learning: Scikit-learn, TensorFlow, PyTorch, XGBoost, CNNs, RNNs, Transformers, Feature Engineering, Model Evaluation

Databases: DynamoDB, PostgreSQL, MongoDB, MySQL, SQLite, SQLAlchemy ORM

Data & Visualization: Pandas, NumPy, Plotly, Streamlit, Matplotlib, Seaborn

Specialized: Multi-tenant RBAC, GST/MSME Compliance, Tally XML, Jira REST APIs, AVIXA Compliance Tooling, Three.js

Certifications

- AWS Certified Solutions Architect – Associate
- Data Engineering with Hadoop & Spark
- Mastering Generative AI & Large Language Models

Education

B.Tech -Artificial Intelligence & Data Science (Honors in Cybersecurity)

A.C. Patil College of Engineering, Navi Mumbai

GPA: 8.64 / 10.0

Aug 2021 -May 2025